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Sarcodonin G Derivatives Exhibit Distinctive Effects on Neurite Outgrowth by Modulating NGF Signaling in PC12 Cells

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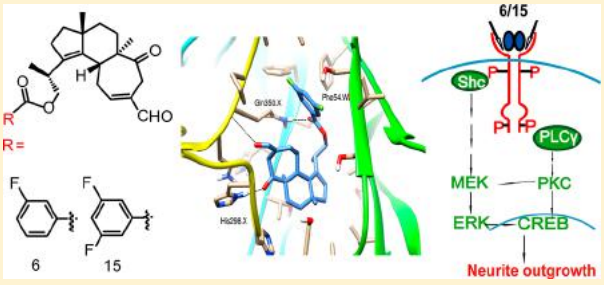
Supporting Information

ABSTRACT: Sarcodonin G, one of the cyathane diterpenoids isolated from the mushroom *Sarcodon scabrosus*, possesses pronounced neurotrophic activity but ambiguous mechanical understanding. In this work, sarcodonin G was chosen as a lead compound to prepare a series of 19-*O*-benzoyl derivatives by semisynthesis and their neuritogenic activities were evaluated. **6** and **15** (10 μM) were investigated with opposite effects in PC12 cells. **6** exhibited a superior activity to sarcodonin G by promoting NGF-induced neurite outgrowth, while **15** showed an inhibitory effect. Supportingly, **6** and **15** (20 μM) significantly induced and suppressed neurite extension in primary cultured rat cortical neurons, respectively. In mechanism, the two derivatives were revealed to influence NGF-induced neurite outgrowth in PC12 cells through the regulation of PKC-dependent and -independent ERK/CREB signaling as well as the upstream TrkA receptor phosphorylation. Furthermore, a possible pattern of interaction among NGF, **6**/**15** and TrkA was presented using molecular simulations. It revealed that **6**/**15** may contribute to the stabilization of the NGF-TrkAd5 complex by establishing several hydrophobic and hydrogen-bond interactions with NGF and TrkA, respectively. Taken together, **6** and **15** modulate PKC-dependent and -independent ERK/CREB signaling pathways possibly by influencing the binding affinity of NGF to the receptor TrkA, and finally regulate neurite outgrowth in PC12 cells.



KEYWORDS: *Cyathane diterpenoid, sarcodonin G derivatives, neurite outgrowth, ERK/CREB pathway, PKC, TrkA*

INTRODUCTION

Cyathane diterpenoids are a typical family of fungal metabolites unified by the presence of a characteristic angularly fused contiguous 5/6/7 tricarbocyclic skeleton with a trans 6/7 ring junction.[1] Accumulating evidence supports the notion that these natural products are bioactive compounds with multiple activities including fascinating neurotrophic properties.[2,3] Erinacines[4,5] and herinacines[6,7] from *Hericium erinaceus*, cyrneines from *Sarcodon cyrneus*,[8−10] glaucopines from *Sarcodon glaucopus*,[11,12] stratoids from *Cyathus striatus*,[13] as well as scabronines[14−18] and sarcodonins[19] from *Sarcodon scabrosus* exhibit stimulatory effects on nerve growth factor (NGF) synthesis or secretion and neuritogenesis in distinct cell lines. This structural and functional diversity suggested their potential as lead compounds for identifying and exploring small molecules with neurotrophic activities. Inspiringly, preliminary structure−activity relationship (SAR) studies implicated a structural feature that can identify the NGF-induced neuritogenic activities in rat pheochromocytoma PC12 cells,[19,20] including 9-methyl (scabronine K vs sarcodonin G), α,β-unsaturated $\Delta^{11,12}$ aldehyde groups (scabronine L, H, and B vs sarcodonin A and G), and the double bond at C-3/C-4 (scabronine M vs sarcodonin G) (Scheme 1). However, little is known about the essential role of the other groups on cyathane carbon skeleton, such as 19-hydroxyl, let alone the detailed mechanism underlying the varying effects of these sarcodonins and scabronines on the NGF-induced neuritogenesis.

In our ongoing efforts to explore more effective neurotrophic compounds,[19−24] the most potent sarcodonin G (G) was chosen as a lead compound to obtain a 19-*O*-acetyl derivative (**1**) by semisynthesis. It was found that this modification improved G's activity (10 μM), indicating the potential of 19-hydroxyl to develop derivatives for optimizing the neurotrophic effects.

In this paper, we further prepared a series of 19-*O*-benzoyl derivatives (**2**−**16**) by semisynthesis and evaluated their neuritogenic activity in PC12 cells as well as primary cultured rat cortical neurons. Two derivatives named **6** and **15**, bearing minor difference in structure but dramatic difference in activity, were screened to identify the essential signaling pathways involved in the regulation of NGF-induced neurite outgrowth. Furthermore, a possible action mode of G compounds was revealed with the aid of microscale thermophoresis (MST)[25] and molecular simulations.

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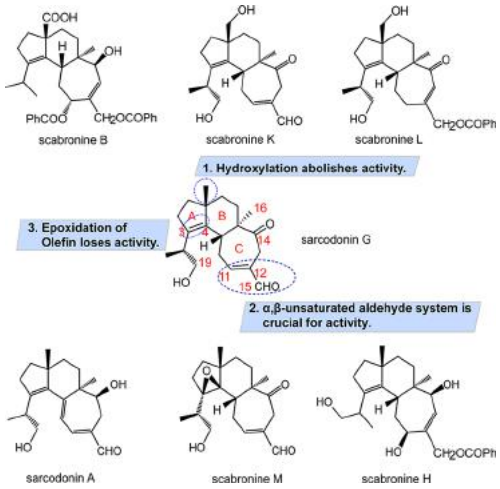
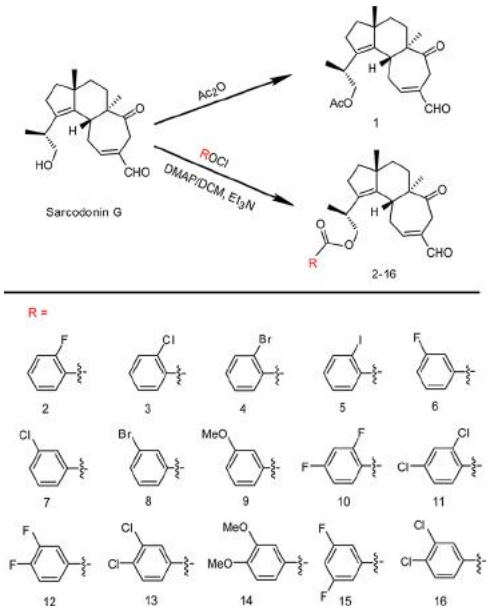
Scheme 1. Structure−Activity Relationship Study of Seven Cyathane Diterpenoids



RESULTS AND DISCUSSION

Semisynthesis of G 19-*O*-Benzoyl Derivatives. G was quantitatively obtained from the fruit bodies of *S. scabrosus.* The procedures are shown in the Supporting Information. Semisynthesis of 19-*O*-acetyl derivative and a series of 19-*O*-benzoyl derivatives is shown in Scheme 2. Briefly, G was treated with

Scheme 2. Semisynthesis of G Derivatives 1−16



dry Ac_2O or respective benzoyl chloride derivatives in the presence of DMAP at room temperature. Workup of the resulting residue provided new compounds 1−16 (purity > 99.0% for HPLC) as yellowish oils in good or excellent yields (Table S1). Their structures were characterized by means of IR, NMR, and HRMS, and the spectroscopic data are listed in the Supporting Information.

Effects of G and Its 19-*O*-Benzoyl Derivatives on NGF-Induced Neurite Outgrowth in PC12 Cells. PC12 cell line was used as a model system for neuronal differentiation studies.[26] We therefore investigated the influence of **1−16** (10 μM) on the NGF-induced neurite outgrowth in PC12 cells. The derivatives affected significantly the cellular morphology in the presence of NGF other than the absence of NGF (Figure 1a). Coadministration of the derivatives and NGF influenced the percentage of neurite-bearing cells with different amplitude compared with the vehicle control (Figure 1b). In case of monohalo substituted 19-*O*-benzoyl derivatives, the *meta*-substituted compounds displayed higher activity than that of *ortho*-substituted ones (e.g., **6** > **2**, **7** > **3**, **8** > **4**). And the derivatives increased the activity with decreasing atomic number (e.g., **2** > **3** > **4** > **5** and **6** > **7** > **8**). Taking into account the 19-*O*-methoxybenzoyl derivatives, the relative activity can be expanded as **6** > **7** > **8** > **9.** Among the 19-*O*-benzoyl derivatives, *meta*-fluoro substituted derivative (**6**) exhibited the highest activity even superior to G ($p < 0.01$). In parallel, **6** upregulated the NGF-induced expression of neuronal differentiation marker proteins (i.e., growth-associated protein 43 (GAP43) and β3-tubulin)[27] compared with G (Figure 1c). On the other hand, *ortho*-iodo substituted derivative (**5**), 3,4-difluoro substituted derivative (**12**), and 3,5-difluoro substituted derivative (**15**) abolished G's activity. The SAR study revealed that the 19-hydroxyl cannot tolerate different modifications, suggesting that the hydroxyl might be critical for binding to its biological targets. Therefore, the proper modification at 19-hydroxyl group of G could optimize its neurotrophic efficacy. Surprisingly, **6** (1−10 μM) evoked a dose-dependent increase in NGF-induced neurite outgrowth (Figure 1d), whereas **15** promoted NGF-induced neurite outgrowth at lower concentration (1−5 μM) and displayed inhibitory effects at higher concentration (10 μM) (Figure 1e). Accordingly, we chose the most impressive derivatives **6** and **15** to explore the underlying mechanism in PC12 cells. None of the derivatives affected the survival or proliferation of PC12 cells below 20 μM (Figure S1). We therefore chose the maximal concentration of 10 μM for subsequent experiments in PC12 cells.

Effects of G and Its 19-*O*-Benzoyl Derivatives on Neurite Outgrowth in Primary Cultured Rat Cortical Neurons. Similar to the case of PC12 cells, **6**/**15** (20 μM) did not affect cell viability (Figure 2b) but did facilitate and inhibit neurite growth in primary cultured rat cortical neurons, respectively (Figure 2a). Particularly, the maximal neurite length for the neurons treated with **6** (5−20 μM) or **15** (0.5−10 μM) increased dose-dependently. However, **15** (20 μM) suppressed neurite outgrowth (Figure 2c). These results not only implied that **6**/**15** also regulated the neurite outgrowth in native neurons but also supported the distinctive effects of **6**/**15** observed in PC12 cells.

Effects of 6 and 15 on the Activation of CREB and Erk1/2 in PC12 Cells. It was known that the activation of cAMP-response element binding protein (CREB) initiated the transcription of genes associated with the neuritogenesis, neurogenesis and cognition function.[28] And (extracellular regulated protein kinases 1/2) Erk1/2 was one of the upstream signalings of CREB.[29] To investigate the role of Erk1/2 and CREB in the regulation of NGF-induced neurite outgrowth by **6** and **15** in PC12 cells, we examined the phosphorylation level of CREB and Erk1/2 (Figure 3a). It showed that **6** evoked a dose-dependent increase in NGF-mediated CREB and Erk1/2 phosphorylation, whereas 10 μM **15** caused decrease. Meanwhile, the blockage of Erk1/2 activation by PD 98059 and U0126[30] abolished the **6**/**15**-potentiated NGF-induced neuritogenesis (Figure 3b). These findings demonstrated that MAPK/ERK cascade was indispensable in the coordination between G derivatives and NGF in CREB-initiated neurite outgrowth of PC12 cells.

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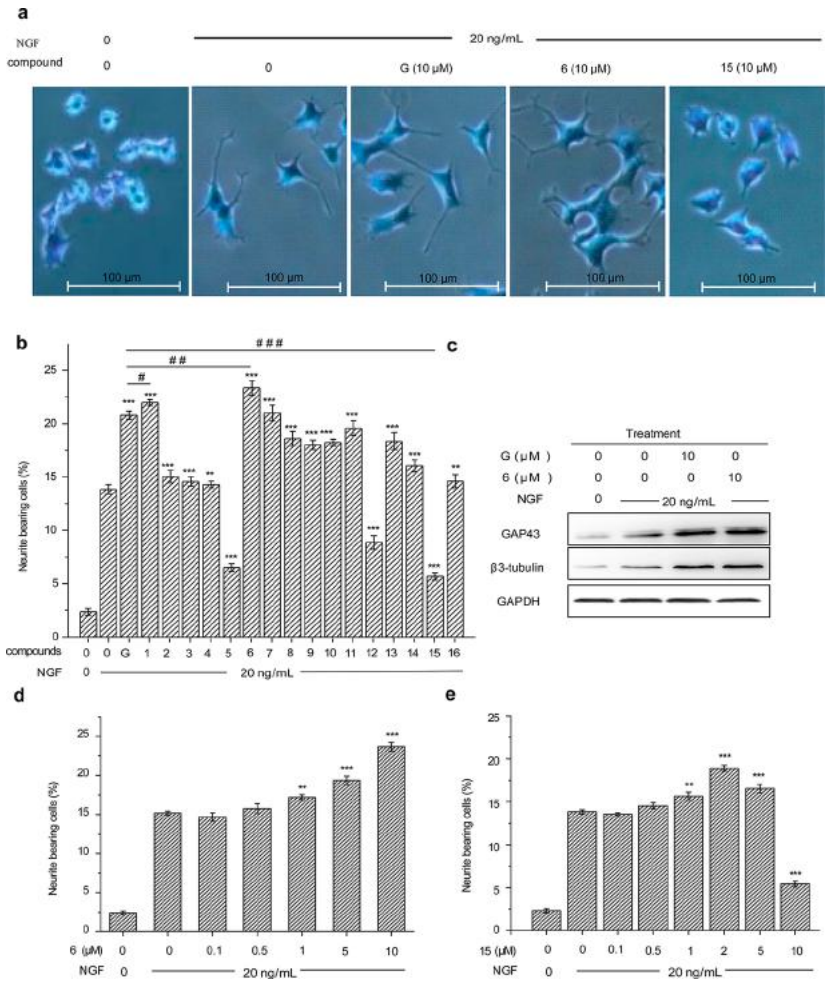

Figure 1. Effects of G and its derivatives on the NGF-induced neurite outgrowth in PC12 cells. PC12 cells were treated with 0.1% DMSO (negative control), 20 ng/mL NGF (positive control), or indicated agents for 48 h as described in the Analysis of Neurite Outgrowth in PC12 Cells. (a) Morphological characteristics of PC12 cells. Scale bar: 100 μm. (b) Percentage of neurite-bearing PC12 cells. Data indicated the mean ± SD (n = 3). $^{**}p < 0.01$ and $^{***}p < 0.001$ represent significant differences versus NGF control. $^{\#}p < 0.05$, $^{\#\#}p < 0.01$, and $^{\#\#\#}p < 0.001$ represent significant differences versus G group (ANOVA followed by Dunnet's posthoc test). (c) Western blotting analysis of the protein expression of neuronal differentiation markers: β3-tubulin and GAP43. GAPDH was detected as the control of protein loading. (d, e) Dose−effect relationship of **6** and **15**, respectively. Values indicated the mean ± SD (n = 3). $^{**}p < 0.01$ and $^{***}p < 0.001$ represent significant differences versus NGF control (ANOVA followed by Dunnet's posthoc test).

Effects of 6 and 15 on PKC Mediated ERK/CREB Signaling Pathways in PC12 Cells. Protein kinase C (PKC) has been reported involved in the NGF-induced neurite extension in PC12 cells.[29] We therefore evaluated PKC activity in the PC12 cells treated by indicated agents. It revealed that **6** increased significantly the NGF-induced PKC activity in a dose-dependent manner. However, 10 μM **15** decreased modestly the NGF-induced PKC activity (Figure 4a). Moreover, the blockage of PKC impaired partially but not completely the **6**-promoted neuritogenic activity of NGF (Figure 4b) as well as Erk1/2 and CREB phosphorylation (Figure 4c,d). Considering ERK's essential role in the coordination between G derivatives and NGF in the CREB-initiated neurite outgrowth of PC12 cells, it can be concluded that the coexistence of PKC-independent and PKC-dependent ERK/CREB signaling pathways in the regulation of **6**/**15** on the NGF-induced neurite outgrowth in PC12 cells.The findings agreed well with previous studies on the cross-talk between PKC and Raf/MEK/ERK cascade.[31−33]

To be more specific, the similar inhibitory effects on the percentage of neurite-bearing cells by GF109203X (selective inhibitor for PKCα, βI, βII, γ) and Enzastaurin (selective inhibitor for PKCβI, βII) distinguished PKCβ as the possible functional one. In fact, the other PKC isoforms including PKCδ,[34] PKCε,[30,35] PKCζ,[35] and PKCι,[36] were shown to be required for NGF-induced differentiation. However, the essential role of PKCδ has been questioned since it was based on the inhibitory effects of rotterlin which was further identified as an antagonist for numerous kinases other than a selective PKCδ inhibitor.[37] In contrast, both GF109203X and Enzastaurin possessed specific antagonistic action on their respective PKC isoforms. Therefore, it can be inferred that sarcodonin G derivatives may act as a regulator of PKCβ to influence the NGF-induced neuritogenesis, consistent with a concomitant accumulation of PKCβII in the potentiation of NGF-mediated neuronal differentiation by H7 in PC12 cells.[38]

Effects of 6 and 15 on TrkA Activation in PC12 Cells. NGF initiates its signaling transduction by binding two receptors: tyrosine kinase receptor A (TrkA)[39] and p75 neurotrophin receptor ($p75^{NTR}$).[40] In response to NGF, TrkA induces receptor dimerization, autophosphorylation and a cascade of signaling events, including activation of Shc, phospholipase C-γ (PLCγ), and phosphatidylinositol 3-kinase (PI3K), followed by

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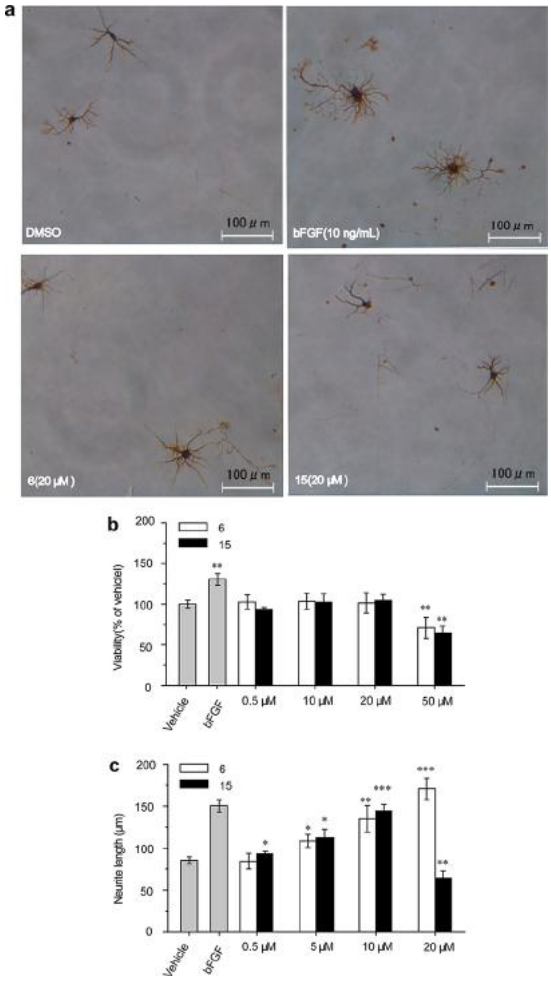

Figure 2. Effects of G and its derivatives on the NGF-induced neurite outgrowth in primary cultured rat cortical neurons. The experiments were conducted as described in the Analysis of Neurite Outgrowth in Primary Cultured Cortical Neurons section. (a) Cell morphology was observed and photographed. Scale bar: 100 μm. (b) Cell viability of primary rat cortical neurons (% of vehicle). Values indicate the mean ± SD (n = 6). $^{**}p < 0.01$ represents significant differences versus vehicle (ANOVA followed by Dunnet's posthoc test). (**c**) Neurite length of primary rat cortical neurons. Data are expressed as the mean ± SD (n = 3). $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$ represent significant differences compared with vehicle (ANOVA followed by Dunnett's test).

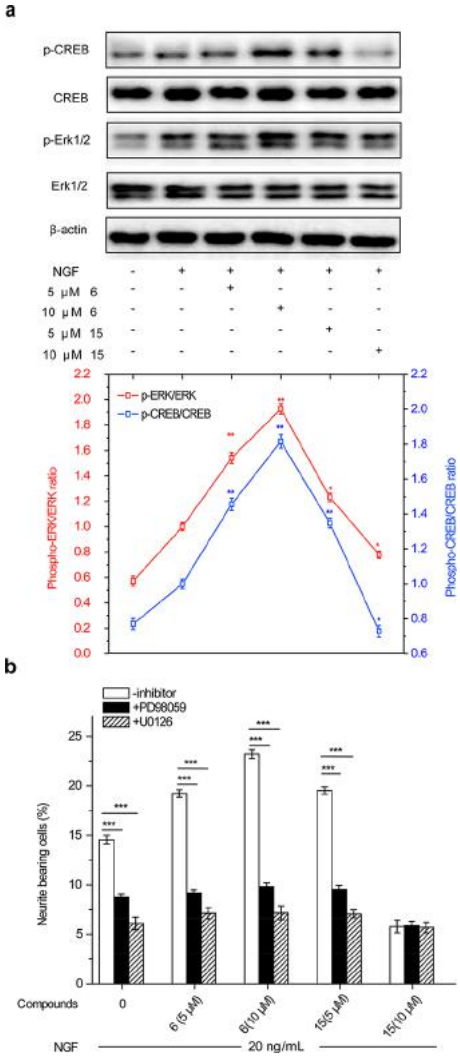

Figure 3. Effects of **6**/**15** on the CREB and ERK signaling pathways in PC12 cells. (a) Phospho-protein and protein were analyzed by Western blotting as described in Methods. The cells were treated with NGF and **6**/**15**, alone or in combination for 30 min. β-Actin was detected as the control of protein loading. Normalized intensity of phospho-protein versus protein was presented as the mean densitometric value ± SD (n = 3). $^{*}p < 0.05$ and $^{**}p < 0.01$ represent significant differences compared with NGF control. (ANOVA followed by Dunnet's posthoc test.) (**b**) Effects of inhibitors on the percentage of neurite-bearing cells. PC12 cells were treated by MEK inhibitors (PD 98059, 20 μM and U0126, 5 μM) for 1 h prior to 48 h exposure to indicated agents. Data are expressed as the mean ± SD (n = 3). $^{*}p < 0.05$ and $^{***}p < 0.001$ represent significant differences compared with inhibitor-free group (Student's t test).

initiation of the Ras-ERK, PKC, and Akt, respectively. These cascades converge signal to CREB-mediated gene expression associated with NGF-induced neurite outgrowth in PC12 cells.[29] Based on the analysis of signaling pathways in our research, it can be reasonably inferred that G derivatives targeted NGF/TrkA signaling in the regulation of NGF-induced neurite outgrowth in PC12 cells. To evaluate whether G derivatives serve as a NGF potentiator for TrkA activation, PC12 cells were treated with NGF alone or in combination with various drugs for 15 min to analyze the phosphorylation level of TrkA. NGF-induced TrkA phospharylations at Y490 and Y785 increased obviously as **6** increased, whereas decreased markedly as **15** increased (Figure 5a). Considering that Y490 and Y785 phosphorylation are responsible for activation of Ras and PLCγ, respectively,[41] it suggested that G derivatives regulated NGF-induced TrkA phosphorylation at both Y490 and Y785 and thus modulated ERK and PKC signaling cascades, respectively. These results further confirmed that the concomitance of PKC-independent and -dependent ERK activation. Moreover, an efficient tyrosine kinase (Trk) inhibitor K252a (IC_{50} = 3.2 nM for TrkA) suppressed the NGF-induced neuritogenesis dramatically in all experimental groups. And a highly selective TrkA inhibitor GW441756 (IC_{50} = 2 nM)[42,43] abolished the activity more completely than K252a (Figure 5b). These results supported convincing evidence about a central role of TrkA in the stimulatory effects by **6**/**15**. Above all, it can be demonstrated that NGF, as a potent ligand of the receptor TrkA, is indispensable for G compounds to provoke signaling transduction. The stimulatory effect by G derivatives was mainly mediated by NGF-induced

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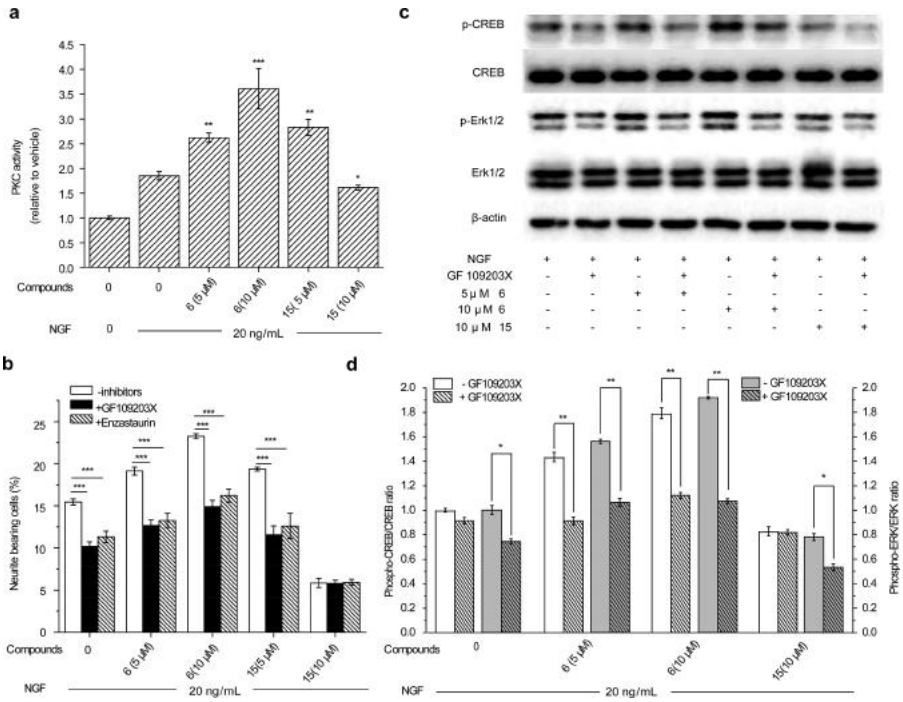

Figure 4. Effects of **6**/**15** on the PKC-mediated ERK/CREB signaling pathways in PC12 cells. (a) Normalized activity of PKC was analyzed by ELISA kit as described in PKC Activity Analysis. Data is presented as the mean ± SD (n = 4). $^{**}p < 0.01$ and $^{***}p < 0.001$ represent significant differences versus NGF control (ANOVA followed by Dunnet's posthoc test). (b) Effects of PKC inhibitors on the percentage of neurite-bearing cells. PC12 cells were pretreated by 2 μM GF109203X or 1 μM Enzastaurin for 1 h followed by 48 h exposure to indicated agents. Data is presented as the mean ± SD (n = 3). $^{***}p < 0.001$ represent significant differences versus inhibitor-free group (Student's t test). (c) Phospho-protein and protein were analyzed by Western blotting as described in Methods. The cells were treated with or without GF 109203X for 1 h prior to incubation by NGF and **6**/**15**, alone or in combination for 30 min. β-actin was detected as the control of protein loading. (d) Densitometric analysis of protein phosphorylation levels on Western blotting (c). Normalized intensity of p-protein versus protein is presented as the mean densitometric value ± SD (n = 3). $^{*}p < 0.05$ and $^{**}p < 0.01$ represent significant differences compared with inhibitor-free group (Student's t test).

TrkA-Y490 and TrkA-Y785 phosphorylation which evoked PKC-independent and PKC-dependent ERK/CREB signaling pathways respectively and finally regulate neurite outgrowth in PC12 cells.

Molecular Models and Interaction Hypothesis. Firstly, MST test excluded the direct binding of our compounds to NGF (Figure S2a).[25] Furthermore, the neurite outgrowth analysis showed that G or **6**/**15** also evoked an increase of gambogic amide (NGF nonpeptidomimetic)-induced neuritogenesis (Figure S2b). Therefore, the neurotrophic activity of G compounds should be mediated by more complicated interaction among NGF, TrkA and the G compounds.

It was known that Ig-like domain (d5) in the extracellular region of the TrkA receptor was essential for binding of NGF with high affinity[44] and the crystal structure of NGF in the complex with d5 of TrkA has been well resolved (PDB code: 1WWW).[45] Thus, molecular docking and molecular dynamics stimulation were performed between **6**/**15** and NGF-TrkAd5 complex. The MD simulations revealed that both **6** /**15** molecules penetrated into the interfaces between NGF and TrkAd5 in a symmetrical pattern (Figure 6a). Macroscopically, **15** mediated tighter binding between NGF and TrkAd5 than **6** (Figure 6a). Microscopically, **15** established more sophisticated interactions with NGF as well as TrkAd5 than **6** (Figure 6b, c), such as $\pi-\pi$ stacking between 19-benzoyl ring of **15** and F54 of TrkA. The interaction sites located within proteins were proved to be vital for the stabilization of the NGF-TrkAd5 complex by the previous MD simulations[46] and biological mutagenesis data of TrkA.[47] In addition, the root-mean squared deviation (RMSD) profiles of the specific binding parts were more flexible for **6** than **15**, underlying the binding of **15** to NGF-TrkAd5 complex may be more stable than **6** (Figure S2c and d). It was consistent with the binding energy comparison between **6** and **15** (−11.68 kcal/mol vs −11.86 kcal/mol) as well as the MD simulations. Taken together, **6**/**15** may provide contributions to the affinity of the complex by acting as a sort of effective connection between NGF and the receptor TrkA. Presumably, the *meta*-substitution displayed superior activity to *ortho*-substitution may be due to their favorable formation of hydrogen bond with the Q350 of TrkAd5 and **6** had the highest neurotrophic activities may be ascribed to its strongest hydrogen bonding capacity. On the other hand, it can be assumed that **15** could establish tighter interaction with the protein complex than **6**. Probably, the kinetics of reversible association/dissociation will be disrupted as the concentration of **15** increases,[48] resulting in the inhibitory effects on the NGF/TrkA signaling.[49]

In conclusion, we demonstrated for the first time two 19-*O*-benzoyl derivatives of G with opposite activities at 10 μM and explored the detailed mechanisms (Scheme 3). **6** and low concentration of **15** promoted NGF-induced neurite outgrowth via increasing phosphorylation levels of the upstream receptor TrkA, up-regulating downstream PKC-independent and -dependent ERK/CREB signaling pathways in PC12 cells. In contrast, high-concentration of **15** inhibited NGF-induced neurite outgrowth via decreasing phosphorylation levels of TrkA, down-regulating PKC-independent and -dependent ERK/CREB signaling pathways. Furthermore, a possible pattern of interaction among NGF, **6**/**15**, and TrkAd5 was presented by

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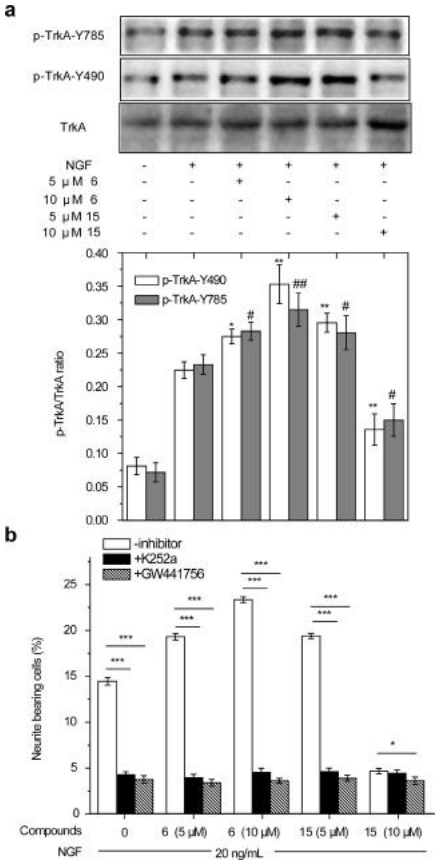


Figure 5. Effects of **6** and **15** on the TrkA signaling pathway in PC12 cells. (**a**) Effects of **6**/**15** on the TrkA phosphorylation in PC12 cells. The cells were treated with NGF and **6**/**15**, alone or in combination for 15 min. Phospho-TrkA (p-TrkA) and TrkA were analyzed by Western blotting as described in Methods. Normalized intensity of p-TrkA versus TrkA is presented as the mean ± SD ($n = 3$). $^{*}p < 0.05$, $^{**}p < 0.01$ and $^{\#}p < 0.05$, $^{\#\#}p < 0.01$ represent significant differences compared with respective NGF controls (ANOVA followed by Dunnet's posthoc test). (b) Effects of selective TrkA inhibitor on the NGF-induced neurite outgrowth in PC12 cells. Values show the percentage of neurite-bearing cells after 1h pretreatment with or without 0.1 μM K252a or GW441756 followed by 48h exposure to indicated agents. Data is presented as the mean ± SD ($n = 3$). $^{***}p < 0.001$ represents significant differences versus inhibitor-free group (Student's t test).

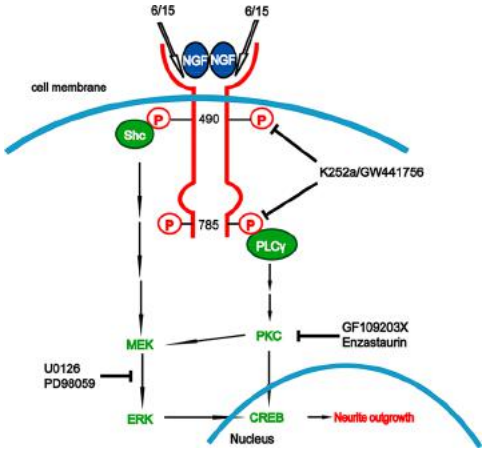
the molecular simulations. It suggested that **6**/**15** could strengthen ligand−receptor binding, which contributed greatly to the stabilization of the NGF-TrkA complex. These explorations will provide beneficial references for the structural optimization of sarcodonins for their potential clinical application to combat neurodegenerative disorders.

■ METHODS

Chemicals and Reagents. Poly-L-lysine (PLL), DMSO, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT), and protein kinase inhibitiors (PD98059, U0126, LY294002, GF109203X, Enzastaurin, and K252a) were purchased from Sigma-Aldrich Co. (St. Louis, MO). GW441756 was purchased from APExBIO Co. (Houston, TX). Gambogic amide was purchased from Abcam Co. (Cambridge, UK). The mouse 7S NGF was purchased from Wuhan Hitech Biological Pharma Co. LTD (Wuhan, China). Human recombinant basic fibroblast growth factor (bFGF) was purchased from PeproTech, Inc. (Rocky Hill, NJ). The primary antibodies were all purchased from Cell Signaling Technology, Inc. (Beverly, MA) .

Cell Culture. PC12 cells were obtained from China Center for Type Culture Collection (CCTCC). The cells were incubated in

Scheme 3. Schematic Presentation of a Possible Mechanism by Which 6/15 Regulates NGF-Induced Neurite Outgrowth[a]



[a]The role of TrkA, ERK, and PKC was investigated using K252a/GW441756, U0126/PD98059, and GF109203X/Enzastaurin, respectively. **6**/**15** modulates NGF-TrkA interaction and thus effectively influences the phosphorylation of TrkA, leading to the adjustments of PKC and ERK signaling pathways. Ultimately, **6**/**15** regulates the activation of CREB which initiates the neuritogenesis in PC12 cells.

nutrient mixture F-12 (Ham) medium (F12, Gibco BRL Co.Ltd.: Gaithersburg, MD) supplemented with 10% heat-inactivated horse serum (HS, Gibco), 5% inactivated fetal bovine serum (FBS, Gibco), 100 U/mL penicillin G, 100 g/mL streptomycin, and 2.5 g/L sodium bicarbonate in humidified air containing 5% CO_2 at 37 °C. Primary rat cortical neurons were separated from the cerebral hemispheres of 18-day fetal Sprague−Dawley (SD) rats (provided by the Experimental Animal Center of Shannxi Province). The cells were cultured in Dulbecco's modified Eagle's medium (DMEM, Gibco) supplement with 10% FBS.

Analysis of Neurite Outgrowth in PC12 Cells. PC12 cells were seeded into PLL-coated 24-well plates at a density of 2×10^4 /mL with normal serum medium for 24 h, and then treated with low serum (2% HS and 1% FBS) medium for 14 h prior to exposure to vehicle (0.1% DMSO) or indicated reagents for indicated periods. The analysis of neurite outgrowth was performed as reported.[19,23]

Analysis of Neurite Outgrowth in Primary Cultured Cortical Neurons. The neurons were seeded into PLL-coated 24-well plates at a density of 5×10^4 /mL. After 24 h, the medium was replaced with Neurobasal medium containing 2% B27 supplement with indicated compounds or bFGF (10 ng/mL) as a positive control. The measurement of neurite length was conducted as described previously.[24,50]

Western Blotting Analysis. PC12 cells (1×10^6 /well) were seeded in PLL-coated 6-well plates in normal serum medium for 24 h, then shifted to low serum medium for 14 h prior to exposure to vehicle or indicated reagents. Total proteins were isolated from the combined lysates of three wells. Protein extraction, SDS−PAGE and immunoblot analysis were conducted as described.[24]

PKC Activity Analysis. A PKC activity assay kit (Enzo Life Sciences, Farmingdale, NY) was used to evaluate PKC activity. PC12 cells (1×10^6/well) were seeded into 6-well plates in normal serum medium for 24 h, then changed to low serum medium for 14 h prior to exposure to drugs for 15 min. An amount of 10 ng of crude protein/30 μL from cell lysates or PKC standard (10 ng) was used to measure the PKC activity per assay according to the manufacturer's instructions. The results were read at 450 nm in a microplate reader.

Molecular Simulation. The 3D crystal structure of NGF-TrkAd5 complex (PDB code: 1WWW) was obtained from the RCSB Protein Data Bank. The standard 3D structure (PDB format) of **6**/**15** was constructed by using the "SKETCH" option function in SYBYL-X, whose configuration was determined by its spectroscopic data. Molecular docking simulations were performed using the software Autodock4.2 Vina along with AutoDockTools (ADT 1.5.6) using the

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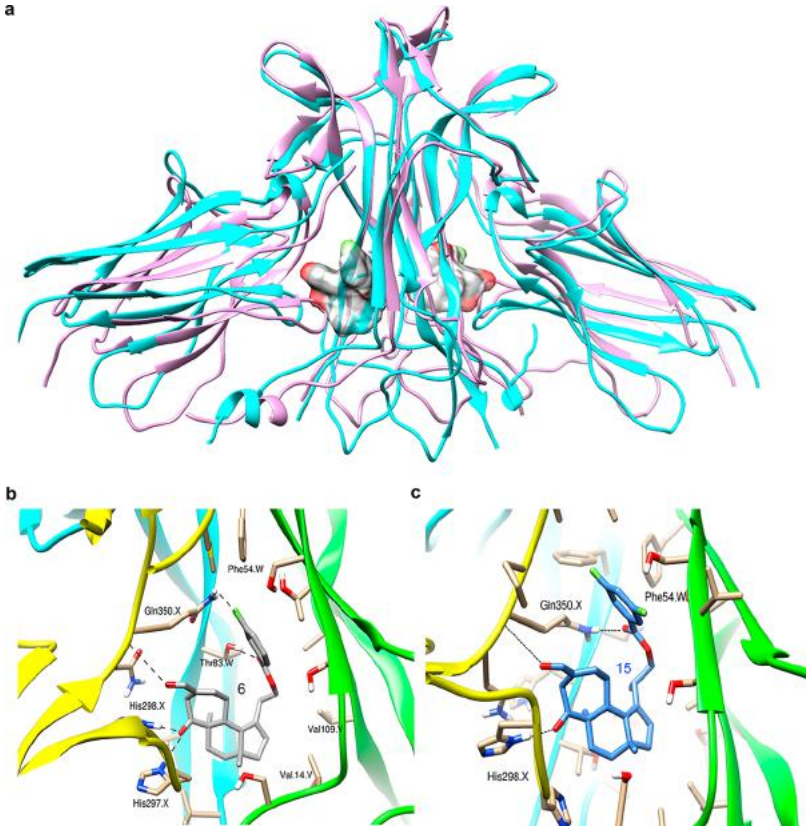

Figure 6. Studies in the molecular understanding of **6**/**15** as NGF potentiators to enhance TrkA-induced neuritogenesis. (a) Molecular docking simulations obtained an averaged conformation of the TrkAd5-NGF-**6**/**15** complex derived from the trajectories of the last converged 500 ps. The TrkAd5-NGF complexes are cyan for **6** and plum for **15**. The molecular surface of **6**/**15** is shown by Chimera. (b, c) **6**/**15** established several interactions with the TrkAd5-NGF complex. **6**/**15** is displayed in stick model and the binding conformation of **6**/**15** is screened from Autodock. The NGF dimmers are green for V-chain and blue for W-chain; TrkAd5 homodimmers are both yellow for X and Y chains. The hydrogen bonds between **6**/**15** and the protein complex are highlighted by black dotted lines.

hybrid Lamarckian Genetic Algorithm (LGA) as our previous study.[51] The lowest binding energy docking conformation was chosen as the initial conformation for a 5000 ps molecular dynamic (MD) simulation by an AMBER 16 package as described before.[51]

Statistics. The results are expressed as mean ± SD. Statistical analyses were conducted using Origin 10.0. Student's *t* test and one-way ANOVA followed by Dunnet's posthoc test were used to compare the differences for groups. Difference was considered significant at p value <0.05.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acschemneuro.7b00488.

Spectroscopic data of **1**−**16** and related experimental procedures (PDF)

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Author Contributions

J.M.G. conceived and designed the experiments; X.W.S. contributed reagents/materials; C.C.Z., C.Y.C., X.Y., and W.C. performed the biological experiments; D.L. executed MD simulations; C.Y.C. analyzed the data; and J.M.G. and C.Y.C. wrote the paper.

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Notes

The authors declare no competing financial interest.

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